

Math 4441: Probability and Statistics

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Chapter 4
Conditional Distributions, Conditional Expectations and Independence
(Part 2 – Continuous Conditional Distributions)
Lectures 22

Math 4441: Probability and Statistics

References:

1. Goodman and Yates – Introduction to Probability and Stochastic Processes, 3rd Edition
2. Scott Miller and Donald Childers – Probability and Random Processes, 2nd Edition
3. Bertsekas and Tsitsiklis – Introduction to Probability, 2nd Edition

Introduction

- A conditional distribution is a complete probability model, and it follows the axioms of probability
- It is called conditional distribution because it is conditioned on the availability of some partial information
- One of the most important topics in probability, because
 - We are often interested in probabilities or expectations with some partial information available
 - It is often useful to assume the availability of some partial information

Conditional distributions extend the conditional probability explained in Chapter 1 for Random variables

Conditional probabilities

Conditional probability

$$P[A|B] = \frac{P[AB]}{P[B]}, P[B] > 0$$

Product Rule or Law of Multiplication

$$P[AB] = P[A|B]P[B] = P[B|A]P[A]$$

Sum Rule or Law of Total Probability

$$P[A] = \sum_{i=1}^n P[A|B_i]P[B_i]$$

Bayes Theorem

$$P[B_i|A] = \frac{P[A|B_i]P[B_i]}{\sum_{i=1}^n P[A|B_i]P[B_i]}$$

Independence

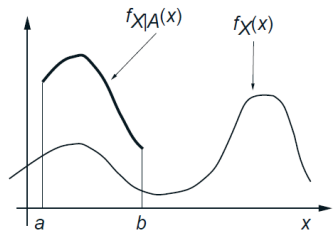
$$P[AB] = P[A]P[B]$$

Conditional Distribution

Conditioned on an Event

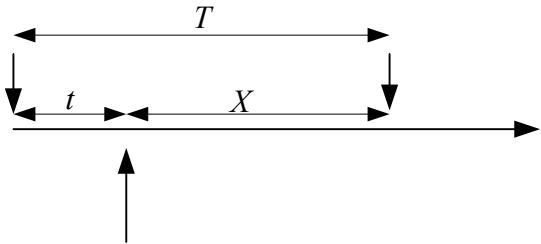
Definition 6.11. For a continuous random variable X with PDF $f_X(x)$ and an event $A \in S_X$ with $P[A] > 0$, where S_X represents the range of values of X , the conditional PDF of X conditioned on A is given by

$$f_{X|A}(x) = \begin{cases} \frac{f_X(x)}{P[A]}, & x \in A; \\ 0, & \text{otherwise.} \end{cases}$$



$$\begin{aligned} F_{X|A}(x) &= P[X \leq x|A] \\ &= \int_{-\infty}^x f_{X|A}(x) dx. \end{aligned}$$

Example 6.20 (The exponential random variable is memoryless). Alvin goes to a bus stop where the time T between two successive buses has an exponential PDF with parameter λ . Suppose that Alvin arrives t seconds after the preceding bus arrival and let us express this fact with the event $A = \{T > t\}$. Let X be the time that Alvin has to wait for the next bus to arrive. What is the conditional CDF $F_{X|A}(x)$?



$$\begin{aligned}
 P[X > x|A] &= P[T > t + x|T > t] \\
 &= \frac{P[T > t + x \text{ and } T > t]}{P[T > t]} \\
 &= \frac{P[T > t + x]}{P[T > t]} \\
 &= \frac{\exp^{-\lambda(t+x)}}{\exp^{-\lambda t}} \\
 &= \exp^{-\lambda x}
 \end{aligned}$$

Example 6.21. Suppose the duration T (in minutes) of a telephone call is an exponential (1/3) random variable with the following PDF

$$f_T(t) = \begin{cases} (1/3)e^{-t/3}, & t \geq 0; \\ 0, & \textit{otherwise}. \end{cases}$$

For calls that last at least 2 minutes, what is the conditional PDF of the call duration?

$$\begin{aligned} f_{T|T>2}(t) &= \begin{cases} \frac{f_T(t)}{P[T>2]}, & t \geq 2; \\ 0, & \textit{otherwise}, \end{cases} \\ &= \begin{cases} \frac{1}{3}e^{-(t+2)/3}, & t \geq 2; \\ 0, & \textit{otherwise} \end{cases} \end{aligned}$$

$$P[A] = P[T > 2] = \int_2^\infty f_T(t)dt = \int_2^\infty \frac{1}{3}e^{-t/3}dt = \frac{1}{3} \left(\frac{-3}{1} \right) e^{-t/3} \Big|_2^\infty = e^{-2/3}$$

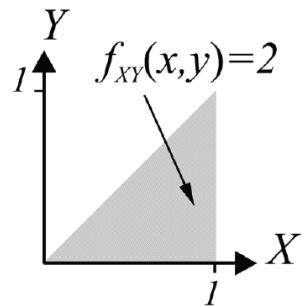
$$\begin{aligned} f_{T|T>2}(t) &= \frac{f_T(t)}{P[T>2]}, t \geq 2 \\ &= \frac{1}{3}e^{-\frac{t-2}{3}}, t \geq 2 \end{aligned}$$

Conditioned on another Random Variable

Definition 6.12. If X and Y have a joint probability density function $f_{XY}(x, y)$, then the *conditional probability distribution* of X , given that $Y = y$, is defined for all values of y such that $f_Y(y) > 0$, by

$$f_{X|Y}(x|y) = \frac{f_{XY}(x, y)}{f_Y(y)}.$$

Example: Suppose random variables X and Y are jointly uniformly distributed within the region $0 \leq y \leq x \leq 1$. Find the conditional PDFs $f_{X|Y}(x|y)$ and $f_{Y|X}(y|x)$.



$$f_X(x) = \int_{-\infty}^{\infty} f_{XY}(x, y) dy = \int_0^x 2 dy = 2x$$

The conditional PDF of Y given X is

$$f_{Y|X}(y|x) = \frac{f_{XY}(x, y)}{f_X(x)} = \begin{cases} 1/x, & 0 \leq y \leq x; \\ 0, & \text{otherwise.} \end{cases}$$

$$f_Y(y) = \int_{-\infty}^{\infty} f_{XY}(x, y) dx = \int_y^1 2 dx = 2(1 - y).$$

Therefore, the conditional PDF of X given Y is

$$f_{X|Y}(x|y) = \frac{f_{XY}(x, y)}{f_Y(y)} = \begin{cases} 1/(1 - y), & y \leq x \leq 1; \\ 0, & \text{otherwise.} \end{cases}$$

Product Rule or Law of Multiplication

$$\begin{aligned}f_{XY}(x, y) &= f_{X|Y}(x|y)f_Y(y) \\ &= f_{Y|X}(y|x)f_X(x)\end{aligned}$$

Example 6.23. First, a point Y is selected at random from the interval $(0,1)$. Then another point X is chosen at random from the interval $(0,Y)$. Find the probability density function of X .

$$f_Y(y) = \begin{cases} 1, & 0 < y < 1; \\ 0, & \textit{otherwise}. \end{cases}$$

$$f_{X|Y}(x|y) = \begin{cases} 1/y, & 0 < y < 1, 0 < x < y; \\ 0, & \textit{otherwise}. \end{cases}$$

$$f_{XY}(x,y) = f_{X|Y}(x|y)f_Y(y) = \frac{1}{y} \cdot 1 = \frac{1}{y}$$

Sum Rule or Law of Total Probability

$$f_X(x) = \sum_i f_{X|A_i}(x)P[A_i]$$

$$\begin{aligned} f_X(x) &= \int_{-\infty}^{\infty} f_{XY}(x, y) dy \\ &= \int_{-\infty}^{\infty} f_{X|Y}(x|y) f_Y(y) dy \end{aligned}$$

Example 6.24. Assume that random variable X represents the voltage at the receiver in a a modem. When symbol “0” is transmitted (event A_0), X is Gaussian $(-5, 2)$ random variable. When symbol “1” is transmitted (event A_1), X is the Gaussian $(5, 2)$ random variable. Given that symbols “0” and “1” are equally likely to be sent, what is the PDF of X ?

$$P[A_0] = P[A_1] = \frac{1}{2}.$$

$$f_{X|A_0}(x) = \frac{f_X(x)}{P[A_0]} = \frac{1}{2\sqrt{2\pi}} e^{-\frac{(x+5)^2}{8}}$$

$$f_{X|A_1}(x) = \frac{f_X(x)}{P[A_1]} = \frac{1}{2\sqrt{2\pi}} e^{-\frac{(x-5)^2}{8}}$$

$$\begin{aligned} f_X(x) &= f_{X|A_0}(x)P[A_0] + f_{X|A_1}(x)P[A_1] \\ &= \frac{1}{4\sqrt{2\pi}} \left(e^{-(x+5)^2/8} + e^{-(x-5)^2/8} \right) \end{aligned}$$

Example 6.25. The PDF of random variable X and the conditional PDF of random variable Y given X are

$$f_X(x) = \begin{cases} 3x^2 & 0 \leq x \leq 1, \\ 0 & \text{otherwise,} \end{cases}$$

and

$$f_{Y|X}(y|x) = \begin{cases} 2y/x^2 & 0 \leq y \leq x, 0 < x \leq 1 \\ 0 & \text{otherwise.} \end{cases}$$

Find the joint PDF of X and Y , and the marginal PDF of Y .

$$f_{XY}(x, y) = f_{Y|X}(y|x)f_X(x) = \begin{cases} 6y & 0 \leq y \leq x, 0 \leq x \leq 1 \\ 0 & \text{otherwise} \end{cases}$$

$$\begin{aligned} f_Y(y) &= \int_y^1 f_{Y|X}(x|y)f_X(x)dx \\ &= \int_y^1 \frac{2y}{x^2} \cdot 3x^2 dx \\ &= 6xy \Big|_y^1 \\ &= 6y(1 - y) \end{aligned}$$

Conditional Expectation

$$E[X|A] = \int_{-\infty}^{\infty} x f_{X|A}(x) dx$$

$$E[g(X)] = \int_{-\infty}^{\infty} g(x) f_{X|A}(x) dx$$

Example 6.26. Find the conditional expectation $E[T|T > t]$ for the problem in Example 6.21.

$$f_T(t) = \begin{cases} (1/3)e^{-t/3}, & t \geq 0; \\ 0, & \text{otherwise.} \end{cases}$$

$$\begin{aligned} f_{T|T>2}(t) &= \frac{f_T(t)}{P[T>2]}, t \geq 2 \\ &= \frac{1}{3}e^{-\frac{t-2}{3}}, t \geq 2 \end{aligned}$$

$$\begin{aligned} f_{T|T>2}(t) &= \begin{cases} \frac{f_T(t)}{P[T>2]}, & t \geq 2; \\ 0, & \text{otherwise,} \end{cases} \\ &= \begin{cases} \frac{1}{3}e^{-(t+2)/3}, & t \geq 2; \\ 0, & \text{otherwise.} \end{cases} \end{aligned}$$

$$E[T|T > t] = \int_2^\infty t \frac{1}{3} e^{-(t-2)/3} dt$$

Integration by parts yields

$$E[T|T > 2] = -te^{-(t-2)/3} \Big|_2^\infty + \int_2^\infty e^{-(t-2)/3} dt = 2 + 3 = 5$$

$$E[X|Y = y] = \int_{-\infty}^\infty x f_{X|Y}(x|y) dx.$$

$$E[g(X, Y)|Y = y] = \int_{-\infty}^\infty g(x, y) f_{X|Y}(x|y) dx$$

Example 6.27. For random variables X and Y the conditional PDF is given by

$$f_{X|Y}(x|y) = \frac{f_{XY}(x, y)}{f_Y(y)} = \begin{cases} 1/(1-y) & y \leq x \leq 1, \\ 0 & \text{otherwise.} \end{cases}$$

Find the conditional expected value $E[X|Y = y]$

$$\begin{aligned} E[X|Y = y] &= \int_{-\infty}^{\infty} x f_{X|Y}(x|y) dx \\ &= \int_y^1 \frac{1}{1-y} x dx \\ &= \left. \frac{x^2}{2(1-y)} \right|_{x=y}^{x=1} \\ &= \frac{1+y}{2}. \end{aligned}$$

Example 6.27. For random variables X and Y the conditional PDF is given by

$$f_{X|Y}(x|y) = \frac{f_{XY}(x, y)}{f_Y(y)} = \begin{cases} 1/(1-y) & y \leq x \leq 1, \\ 0 & \text{otherwise.} \end{cases}$$

Find the conditional expected values $E[X|Y = y]$ and $E[X|Y]$.

Law of Total expectation

$$f_X(x) = \int_{-\infty}^{+\infty} f_{X|Y}(x|y) f_Y(y) dy$$

Example 6.23. First, a point Y is selected at random from the interval $(0,1)$. Then another point X is chosen at random from the interval $(0,Y)$. Find the probability density function of X .

$$Y \sim \text{unif}(0,1)$$

$$X|Y \sim \text{unif}(0,y)$$

$$f_X(x) = \int_{-\infty}^{\infty} f_{X|Y}(x|y) f_Y(y) dy = \int_x^1 \frac{1}{y} dy = \ln y \Big|_{y=x}^{y=1} = \ln 1 - \ln x = -\ln x.$$

$$f_X(x) = \begin{cases} -\ln x, & 0 < x < 1; \\ 0, & \text{otherwise.} \end{cases}$$

Bayes' Theorem for Random Variables

$$f_{X|Y}(x|y) = \frac{f_{Y|X}(y|x)f_X(x)}{\sum_x f_{Y|X}(y|x)f_X(x)}$$

